***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 10 Viral Genomics, Diversity, and Ecology

10.1 Multiple Choice Questions

1) Which type of viruses generally has the smallest genome?
A) bacteriophages
B) DNA viruses
C) RNA viruses
D) viroids
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

2) The Baltimore Scheme to classify viruses contains a total of ________ groups based on ________.
A) four / genome composition
B) four / genome composition and transcription mechanism
C) seven / genome composition
D) seven / genome composition and transcription mechanism
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

3) Early and late viral proteins are classified according to their relative
A) evolutionary appearance in virus genomes.
B) stability during infection.
C) time of synthesis following host infection.
D) transmission into virions.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

4) Which type of viruses can be directly used for translation?
A) dsRNA
B) negative ssRNA
C) positive ssRNA
D) retroviruses
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

5) If the hypothesis stating viruses evolved prior to living organisms on Earth is TRUE, the first type of viruses in the world were likely
A) bacteriophages.
B) DNA viruses.
C) retroviruses.
D) RNA viruses.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.2

6) The filamentous DNA phages are unusual, because they
A) are released from the host without the host being lysed.
B) have linear genomes.
C) replicate without a host.
D) are released from the host without being lysed and have linear genomes.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.3

7) How could overlapping genes in a positive ssDNA virus genome be predicted?
A) Convert the positive ssDNA into its complementary ssDNA and search for genes in the negative ssDNA strand for sequences used in more than one predicted gene.
B) Directly search the three frames of the positive ssDNA for genes that have sequences where more than one gene is predicted.
C) Convert the positive ssDNA into negative ssDNA and search all six possible frames for genes that use part of the same sequence.
D) Convert the positive ssDNA into its complementary ssDNA and search for genes in the negative ssDNA strand that also share a complementary gene in the positive strand.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.3

8) How are T7 genes transcribed?
A) Host RNA polymerase is modified to recognize the T7 promoter.
B) Host RNA polymerase directly translates the T7 genes.
C) T7 has its own RNA polymerase, which is packaged in its capsid and injected into the host during infection to transcribe T7 genes.
D) T7 has its own RNA polymerase, which must first be synthesized by the host.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

9) In T7, the proteins that inhibit the host restriction system are synthesized
A) before the entire T7 genome enters the cell.
B) while the T7 genome is entering the cell but before it enters the nucleus.
C) after the T7 genome is completely within the host cytoplasm.
D) in response to the T7 genome binding to the host chromosome.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

10) A concatemer is a
A) combination of two or more repeated nucleotide sequences covalently linked together.
B) complex of RNA-specific polymerases found only in bacteriophages.
C) linker molecule that allows several phages to infect one host.
D) polymeric protein.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

11) The phage Mu
A) has a circular genome.
B) repairs mutations in the host genome.
C) replicates by transposition.
D) has a circular genome, repairs host genome mutations, and can replicate by transposition.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

12) Mu is a ________ virus with a ________ tail.
A) ssRNA / filamentous
B) dsRNA / helical
C) ssDNA / filamentous
D) dsDNA / helical
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

13) Integration of Mu DNA into the host genome is essential for
A) lytic growth.
B) lysogenic growth.
C) both lytic and lysogenic growth.
D) neither lytic nor lysogenic growth.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

14) Which feature, if changed, would NOT abolish M13′s utility as a cloning vector?
A) ssDNA genome becoming a dsDNA genome
B) loss of genes that make coat proteins
C) replacing the segment of non-coding DNA in its genome with an indispensible gene
D) switch from lysogenic to lytic lifestyle
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.3

15) Of the phages listed below, which creates mutations in its host genome via transposition?
A) lambda
B) M13
C) Mu
D) T7
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

16) Unusually shaped viruses, such as lemon-shaped and spindle-shaped, have recently been discovered in
A) *Archaea*.
B) *Bacteria*.
C) *Archaea* and *Bacteria*.
D) *Eukarya*.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

17) Reoviruses contain ________ genomes, and their replication occurs within the host′s ________.
A) ssDNA / nucleus
B) dsDNA / nucleus
C) ssRNA / cytoplasm
D) dsRNA / cytoplasm
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.10

18) What will happen if the Mu repressor is NOT synthesized?
A) Genome replication will not be able to occur.
B) It will lyse its host.
C) Mu will improperly synthesize its capsid.
D) Transposition will not be possible.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.4

19) Viruses that infect the hyperthermophilic *Archaea* tend to contain genomes that are composed of
A) ssDNA.
B) dsDNA.
C) ssRNA.
D) dsRNA.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

20) Spindle-shaped viruses have been shown to infect only
A) *Eukarya*.
B) *Bacteria*.
C) *Archaea*.
D) plants.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

21) The ________ has been especially useful for genetic engineering because it is capable of triggering a substantial immune response without causing major adverse health effects.
A) adenovirus
B) polymyxavirus
C) vaccina virus
D) herpesvirus
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

22) In designing a drug to inhibit poxvirus, the compound should localize in the host′s ________ to be most effective.
A) nucleus
B) endoplasmic reticulum
C) cytoplasm
D) Golgi complex
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.6

23) The unconventional dsDNA genome replication mechanism where no lagging strand exists is a hallmark of which group of viruses?
A) adenoviruses
B) coronaviruses
C) herpes viruses
D) pox viruses
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.6

24) The hepadnavirus DNA polymerase acts as which of the following?
A) DNA polymerase
B) reverse transcriptase
C) protein primer for synthesis of a strand of DNA
D) DNA polymerase, reverse transcriptase, and protein primer for DNA synthesis
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.11

25) Blocking polyomavirus SV40′s ability to integrate its genome into host cells would
A) avoid cancer development caused by the virus.
B) increase the rate of transformation.
C) increase the latent period of SV40.
D) switch SV40 into a lytic lifecycle which would be especially harmful to the host cells.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.7

26) Herpes viruses can cause all of the following diseases in humans EXCEPT
A) cancer.
B) chicken pox.
C) cold sores.
D) spongiform encephalopathy.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.7

27) What is unusual about phage MS2 infection of *Escherichia coli*?
A) All proteins are synthesized simultaneously during infection so there are no early and late proteins.
B) It attaches to the host′s pilus rather than the cell′s surface.
C) It enters through a host cell porin.
D) More than one MS2 phage can be present in an individual *E. coli* cell.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.8

28) Based on its function, which type(s) of viruses likely contain(s) a gene encoding for RNA replicase?
A) dsDNA and ssDNA viruses
B) positive ssRNA viruses
C) positive and negative ssRNA viruses
D) ssRNA and ssDNA viruses
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.8

29) Polyproteins made from human viruses such as poliovirus must be ________ in order to yield the required functional units of the virus.
A) able to interact with VPg proteins
B) chemically modified with either glycolation or methylation
C) post-translationally cleaved
D) properly folded into secondary and tertiary structures
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.8

30) What is the purpose of synthesizing a negative strand RNA in positive stranded ssRNA viruses?
A) enable rolling circle amplification of the genome, which requires both strands of RNA
B) enable transcription of genes occurring on both the negative and positive strands of the genome, such as overlapping genes
C) proofreading of the genome to minimize mutations generated by the polymerase being passed onto virion progeny
D) to serve as the complementary template sequence in genome amplification of the positive strand
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.8

31) Among the largest RNA genome viruses are ________ which contain a ________ genome.
A) coronaviruses / dsRNA
B) coronaviruses / positive ssRNA
C) polioviruses / dsRNA
D) polioviruses / positive ssRNA
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.8

32) As a consequence of the immune system in humans recognizing dsRNA as foreign
A) dsRNA viruses rarely infect humans.
B) dsRNA viruses quickly transcribe their genes into mRNA which is insensitive to immune responses.
C) genomes of RNA viruses are often chemically modified to avoid recognition by human immune cells.
D) the genomes of dsRNA viruses must avoid human immune cells during infection, including replicating their genomes within their own nucleocapsids.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.10

33) How do reoviruses increase the translational activity of human ribosomes so that they can rapidly produce viral proteins during infection?
A) They chemically modify the RNA transcripts with methyl caps in a similar manner to normal eukaryotic RNA processing.
B) They keep a ribosome binding site specific to human ribosomes on their genome.
C) They have introns and sometimes extons in their genomes to make their RNA resemble eukaryotic mRNA.
D) They only adhere to and infect metabolically active host cells where protein synthesis is high.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.10

34) The family of reoviruses contain dsRNA genomes use a ________ replication process.
A) conservative
B) semiconservative
C) retroviral
D) rolling circle
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.10

35) Identifying proteases being essential for the replication of a virus would suggest the virus
A) lyses its host following genome replication.
B) contains at least one polyprotein.
C) has a single-stranded RNA genome.
D) uses at least one set of overlapping genes.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.8

36) A drug designed to inhibit reverse transcriptase activity would target
A) coronaviruses and rhabdoviruses.
B) retroviruses.
C) hepadnaviruses and retroviruses.
D) viruses with RNA genomes.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.11

37) In contrast to positive ssRNA viruses such as coronaviruses and polioviruses, the genome of retroviruses
A) lacks genes encoding for tRNA primers.
B) must first integrate into the host′s genome before transcription.
C) is negative ssRNA.
D) lacks ribonuclease activity.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.11

38) Proteins made by a ribosome reading through the stop codon of a transcript without their own discrete ribosome binding sites
A) are thought to be a primitive mechanism to avoid host defenses.
B) appear most abundant in archaeal viruses and relatively uncommon in bacteriophages.
C) suggest a relatively low level of protein product is essential for the virus due to the rare frequency of these events.
D) create opportunities for viruses to make different capsid proteins.
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.11

39) When two different influenza viruses infect the same cell, their segmented genomes can undergo reassortment which will result in
A) antigenic drift.
B) antigenic shift.
C) loss of neuramidase.
D) loss of hemagglutinin.
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.9

40) Which of the following conditions favors a lysogenic life cycle in bacteriophages?
A) Having ssDNA.
B) Having ssRNA.
C) A lack of host bacteria.
D) The presence of abundant hosts.
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.12

10.2 True/False Questions

1) Viruses are known to infect *Bacteria*, but no virus has yet been found that infects *Archaea*.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

2) To date, there is no evidence that RNA viruses infect *Archaea.*
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

3) Genomics analysis of recently isolated viruses indicate that some viruses contain larger genomes than some bacterial genomes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

4) The Baltimore classification scheme is a useful way to categorize viruses based on their host infectivity.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.1

5) Viruses that contain positive-strand genomes do not share genetic elements with other positive-strand genomes.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

6) Varied transcription mechanisms distinguish the different DNA virus Baltimore classes, whereas varied translational mechanisms distinguish the RNA virus Baltimore classes.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.1

7) Despite viruses require a living host′s metabolism to replicate, it remains unclear whether viruses existed before living cells.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.2

8) One hypothesis on the origin of DNA points to RNA viruses evolving a modified nucleotide that is insensitive to ribonucleases.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.2

9) Due to the genetic diversity of viruses and their lack of ribosomal RNA, nucleotide-based phylogeny studies are not applicable to virology.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.2

10) Bacteriophages that have single-stranded genomes are specialized to minimize energy requirements because just one strand is necessary for replication.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.3

11) Nonfilamentous bacteriophages often can escape its host without lysing, whereas filamentous phages normally induce cell lysis once replicated inside their host.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.3

12) By nature of its infectivity, M13 phages can be used in the laboratory to continually propagate a particular DNA sequence inside of *Escherichia coli* by simply culturing infected *E. coli* in LB.
Answer: TRUE
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.3

13) Knowing the genome of Mu bacteriophages now enables researchers to locate where it incorporates into bacterial genomes.
Answer: TRUE
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 10.4

14) Some virus shapes that infect members of *Archaea* are unique from other viruses that infect eukaryotes and bacteria.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

15) Most archaeal viruses identified appear to have DNA genomes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.5

16) Many viruses that infect humans may illicit a strong immune response causing additional harmful effects, so the discovery of a virus that can induce an immune response without causing harm made it attractive for vaccine development.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.6

17) Due to their indispensible role for copying its genome, an intracellular host protease that attacks the adenoviral protein ends would likely result in halting its replication.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.6

18) In order to replicate its genome, a positive-strand RNA virus must produce a complete negative-strand RNA molecule that serves as the template for protein synthesis.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.9

19) A bacteriophage that lacks its proteinaceous capsid structure is also called a viroid.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.15

20) Viroids infect only fungi.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.15

10.3 Essay Questions

1) Describe how bacteriophages influence the oceans' bacterial populations and nutrient cycling.
Answer: Phages are responsible for killing up to 50% of the marine bacterial population each day. They also play a major role in horizontal gene transfer due to their ability to package DNA from one host and integrate the DNA it carries into other host bacterial cells. With cellular constituents released that would otherwise not be available for other organisms, cycling of these nutrients might be expected to occur at a quicker rate. Another potential hypothesis could be that nutrients are consumed during phage replication that other bacteria might use and are no longer available to the bacteria by instead being incorporated into phage particles.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.12

2) Describe the hypothesis of viruses occurring before living cells and how DNA might have evolved. What is the current hypothesis about the evolutionary relationships between RNA, DNA, viruses, and cellular life?
Answer: One hypothesis on the origin of viruses states they developed prior to the living cells they infect. RNA viruses likely were the first to come about, and when host cells evolved ribonucleases to combat infection by RNA viruses, DNA might have evolved in viruses and a way to circumvent this defense. Because DNA is more stable than RNA, its evolution would also be advantageous and maintained in the cellular hosts.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.2

3) Is there a certain type of virus morphology that is especially known to cause disease in humans? Explain your reasoning.
Answer: No, there is not a particular morphology of viruses that is known to cause diseases in humans. Viral proteins often cause sicknesses, and certain viral morphologies are not strongly correlated with particular proteins. Only one viral morphology has not been described in animal (human) viruses, which is the head and tail morphology typical of bacteriophages.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.11

4) Describe one use of bacteriophage Mu for a bacterial geneticist, and explain why it is useful.
Answer: Answers will vary, but bacteriophage Mu integrates into its host′s genome and when doing so induces mutations. Bacterial geneticists can use Mu to create a library of mutations, which is an especially useful technique in reverse genetics approaches.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.4

5) Explain why the viral genome of the MS2 phages can be immediately translated. What type of genome must it have for this to be the case?
Answer: The MS2 phage has a positive ssRNA genome. Once the genome enters into the host cell, the host′s RNA polymerase can directly bind to the mRNA transcript-like genome to translate the genome into proteins.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.8

6) Why is so much emphasis placed on the genomic composition (e.g., ssRNA, dsDNA) of individual viruses? Provide examples to support your explanation.
Answer: Answers will vary but should highlight the difference of viral genome diversity in composition compared to the three domains of life which are dominated by dsDNA genomes. Because different viruses genomes composition is so diverse, where some are positive ssRNA, negative ssRNA, dsRNA, positive ssDNA, and dsDNA, fundamental molecular mechanisms are also diverse. The genome composition often dictates which proteins must be carried and not made by the host, the type of genome replication mechanism, and the way transcripts are synthesized.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.1

7) Why are phylogenetic studies of viruses more challenging than *Bacteria*? Explain how genes are selected in viruses for phylogeny and the constraints those create.
Answer: Viruses lack ribosomal RNA gene sequences present in *Bacteria* that are commonly used for phylogenetic studies. A universal gene in all viruses is absent, which means only subgroups are used for phylogenetic studies where the gene(s) is/are present. In a similar way as rRNA genes, phylogenetic genes selected in viruses often target essential functions for a particular group of viruses such as capsid proteins. Because no single gene is essential in viruses, whichever gene or genes selected constrains the phylogenetic study to only a subset of viruses.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 10.2

8) Defend why the discovery of prions and viroids changes our view on what it takes to be an infectious particle. Be sure to explain the feature of each that distinguishes them from traditional viruses.
Answer: Traditionally viruses are thought of as protein-encapsulated genetic elements that exclusively replicate inside a living cell yet prions and viroids each lack one of these characteristics. While neither is a true virus, both are infectious particles, so it is clear a diversity of infectious particles exist. Prions lack nucleotide sequences altogether but one could also argue the protein conformation PrP^{SC} itself serves as the blueprint to make available other PrP^{SC} misfolded proteins. The gene for the prion, present in host cells, encodes for PrP^{C} which is functionally different than its propagative and infectious form. Viroids on the other hand lack protein in their structure and their RNA genomes also do not encode for proteins. Viroids being infectious exclusively as RNA, and prions being infectious exclusively as protein expands our knowledge on the minimal requirements to be infectious.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 10.15, 10.16